

What is Ternary Operator with example

A **ternary operator** (also called a **conditional expression**) is a short way to write an `if-else` statement in **one line**.

Formate:

`value_if_true if condition else value_if_false`

Example:

`a = 10`

`b = 20`

`max_num = a if a > b else b`

`print("Max number is:", max_num)`

Write an if-else ladder to categorize age: 0-12: Child 13-19: Teen 20+: Adult. Convert the Age Logic into Python code.

```
age = int(input("Enter age: "))
```

```
if 0 <= age <= 12:
```

```
    print("Child")
```

```
elif 13 <= age <= 19:
```

```
    print("Teen")
```

```
elif age >= 20:
```

```
    print("Adult")
```

```
else:
```

```
    print("Invalid age")
```

Find Leap Year

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int year;
```

```
    cout << "Enter a year: ";
```

```
    cin >> year;
```

```
    // Leap year condition
```

```
    if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
```

```
        cout << year << " is a leap year." << endl;
```

```
    } else {
```

```
        cout << year << " is not a leap year." << endl;
```

```
}  
  
return 0;  
  
}
```

a,b and c are three number. Find the Max Number.

```
#include <iostream>  
using namespace std;
```

```
int main() {
```

```
    int a, b, c;
```

```
    cout << "Enter three numbers: ";
```

```
    cin >> a >> b >> c;
```

```
    int max_num;
```

```
    if (a >= b && a >= c) {
```

```
        max_num = a;
```

```
    } else if (b >= a && b >= c) {
```

```
        max_num = b;
```

```
    } else {
```

```
        max_num = c;
```

```
    }
```

```
    cout << "The maximum number is: " << max_num << endl;
```

```
    return 0;
```

```
}
```